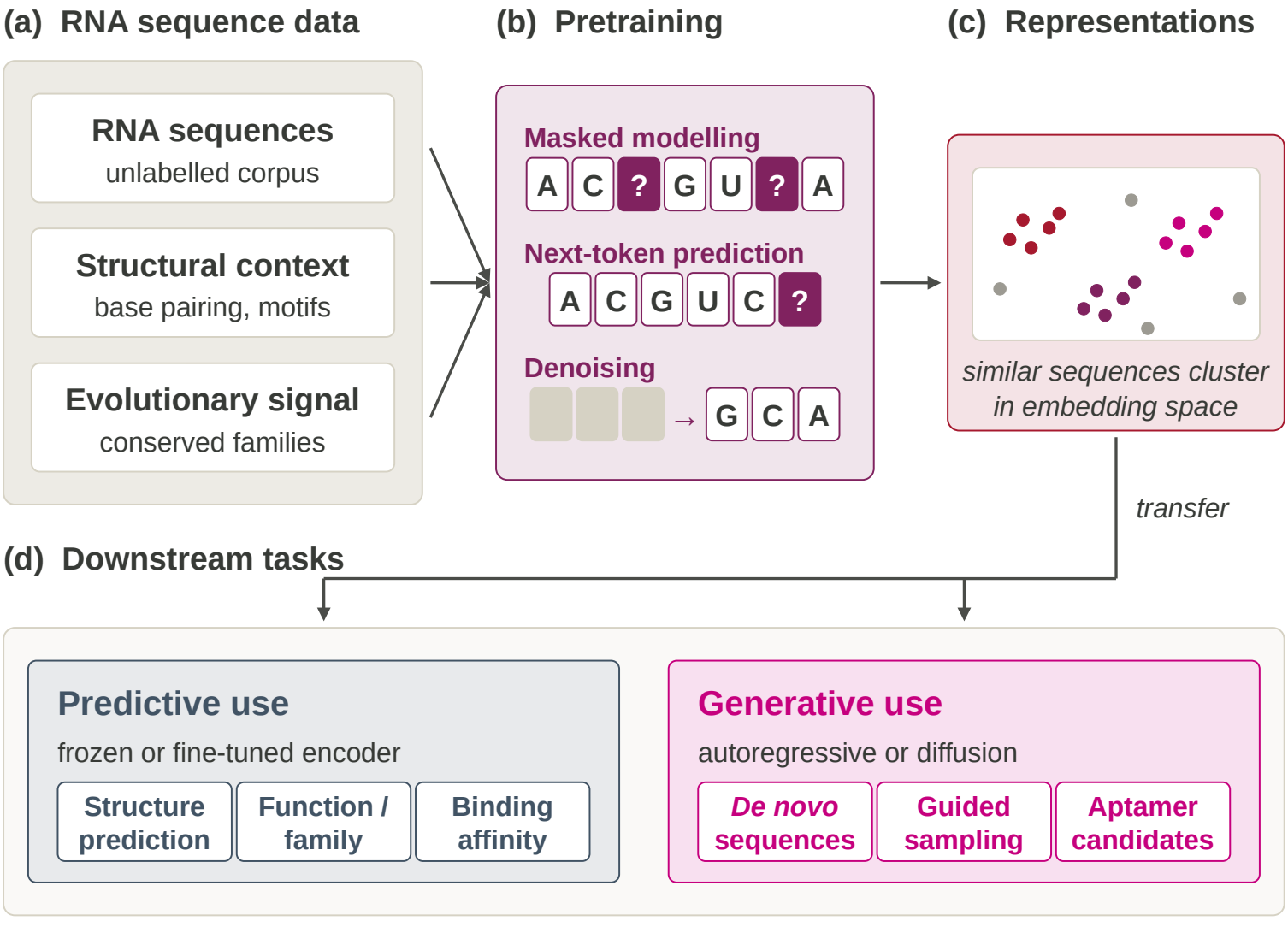

(a) RNA sequence data
RNA sequences
unlabelled corpus
Structural context
base pairing, motifs
Evolutionary signal
conserved families
(b) Pretraining
Masked modelling
A C ? G U ? A
Next-token prediction
A C G U C ?
Denoising
→ G C A
(c) Representations
similar sequences cluster in embedding space
transfer
(d) Downstream tasks
Predictive use
frozen or fine-tuned encoder
Structure prediction
Function / family
Binding affinity
Generative use
autoregressive or diffusion
De novo sequences
Guided sampling
Aptamer candidates